MIL-S-82519(OS)
17 November 1967
Superseding
NOTS XS 512 Rev. N
26 June 1967

MILITARY SPECIFICATION
FOR
SWITCH, SAFETY
MARK 61 MOD 1

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 This specification establishes the requirements of one type of Safety Switch Mark 61 Mod 1. (See 6.1 for intended use and description.)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein except as noted.

SPECIFICATIONS

FEDERAL

PPP-B-621	Boxes, Wood, Nailed and Lock Corner

MILITARY

MIL-P-116	Preservation, Methods of
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible

FSC -1135

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STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging and Packing Levels

MILITARY

MIL-STD-105D	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

DRAWINGS

BUREAU OF NAVAL WEAPONS

LD 540620	Switch, Safety Mark 61 Mod 1 and all documents listed thereon
LD 624603	Tester, Acceptance, Safety Switch

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

OD 13065	Operation and Maintenance of Safety Switch Acceptance Tester for Safety Switch Mark 61 Mod 1

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in the manufacture of the switch shall be in accordance with the requirements specified herein and with the applicable documents listed in BuWeps LD 540620.

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3.2 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the switch shall be manufactured using the methods and procedures proposed for production. The sample shall be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.3 Performance and product characteristics. The switch shall meet the following performance and product characteristics.

3.3.1 Insulation resistance. The switch shall have an insulation resistance of not less than 20 megohms between each lead, and between each lead and the case, when a potential of 500 volts DC is applied.

3.3.2 Acceleration sensitivity. The switch shall remain open while experiencing a constant acceleration of 5g (g = unit of gravitational acceleration) along the sensitive axis. (See Figure 1.)

3.3.3 Switch function. The switch shall close in the time noted for the constant acceleration as follows:

Acceleration (g)	Time (Seconds)
10	0.6 ±0.1
15	0.4 ±0.1

3.3.3.1 Switch latching. The timing weight shall become securely retained by the latch assembly when the switch is functioned, enabling the normally open contacts of the microswitch, an integral part of the switch, to close and remain in a closed condition.

3.3.4 Current capacity and circuit resistance. The voltage drop across the closed circuit of the switch, measured while conducting a continuous current of 2 amperes DC, shall be 0.125 volt DC maximum.

3.4 Environmental requirements. Unless otherwise specified, the switch shall be capable of conforming to the performance and product characteristics of this specification after being exposed to the environments and all conditions specified herein.

3.4.1 Temperature, storage condition. The switch shall withstand temperatures from minus 65 to plus 160 degrees Fahrenheit (F).

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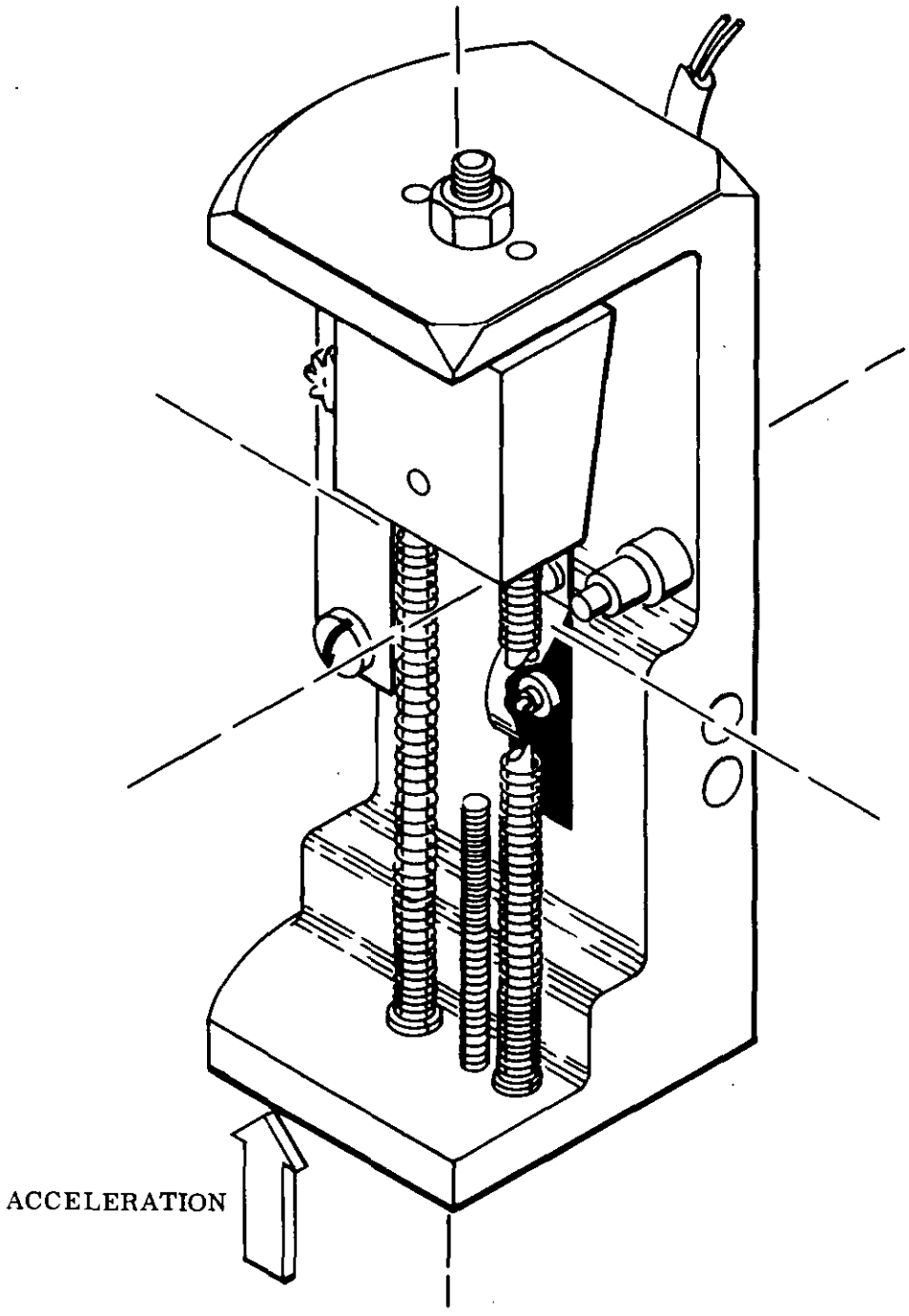


Figure 1. Safety Switch Test Axes for Shock, Vibration and Acceleration

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3.4.2 Vibration, standby condition. The switch shall withstand vibration, as specified for Type I equipment in MIL-STD-167, up to 25 cycles per second (cps).

3.4.3 Shock, standby condition. The switch shall withstand shocks of 40g peak magnitude. The shock shall have a pulse shape approximating a one-half sine wave having a duration of 5 milliseconds.

3.4.4 Temperature, operating condition. The switch shall meet the performance and product characteristics while at temperatures from 35 to 120 degrees F.

3.4.5 Vibration, operating condition. The switch, in the actuated condition, shall withstand vibration through a frequency range of 20 to 2000 cps at 0.10-inch double amplitude or 10g, whichever is less. During vibration, the switch shall meet the current capacity and circuit resistance requirements as specified in 3.3.4.

3.5 Workmanship. The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with the requirements of this specification. All parts used in the assembly shall be thoroughly clean and free of excess materials, chips, and loose or splattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the switch shall be classified as follows:

a. Preproduction and periodic production inspection. (See 4.5.)

b. Quality conformance inspection. (See 4.6.)

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4.3 Sampling. (See 6.2.1.)

4.3.1 Lot size. The term "lot" shall mean "inspection lot", that is a collection of product submitted by a supplier for inspection. Unless otherwise specified in the contract or order the number of product in each lot shall be three hundred (300).

4.3.2 Preproduction sample. A preproduction sample of ten (10) switches, manufactured in accordance with 3.2, shall be subjected to the preproduction tests detailed in 4.5. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the switch by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

4.3.3 Periodic production sample. From each succeeding lot of three hundred (300) or less, a random sample of eight (8) switches shall be selected. These sample units shall be subjected to the tests of 4.5. The switches shall be considered satisfactory if they meet all specification and pertinent drawing requirements and withstand the tests specified in 4.5. (See Table I.) When a defective unit is found in any lot, acceptance of all units in the lot shall be withheld and the contractor shall:

a. Determine the cause of the defect.

b. Take the necessary action, with the concurrence of the procuring activity, to correct or eliminate the defect from all units in the lot and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

4.3.4 Quality conformance inspection sampling. Samples shall be selected as specified in 4.6 and shall be used to determine the acceptability of the switch inspected as specified in 4.6.

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Table I. Preproduction and Periodic Production Test Sequence

Test	Paragraph Number	Test Quantities Preproduction	Periodic
Quality Conformance Inspection	4.6*	10	8
Temperature, Storage Condition	4.5.1.1	10	8
Quality Conformance Inspection	4.6*	10	8
Vibration, Standby Condition	4.5.1.2	10	8
Shock, Standby Condition	4.5.1.3	10	8
Quality Conformance Inspection	4.6*	10	8
Temperature, Operating Condition	4.5.1.4	10	8
Low Extreme	4.5.1.4.1	10	8
Quality Conformance Inspection	4.6	10	8
High Extreme	4.5.1.4.2	10	8
Quality Conformance Inspection	4.6	10	8
Vibration, Operating Condition	4.5.1.5	10	8
Current Capacity and Circuit Resistance Test after Functioning	4.6.1.4	10	8

*Except for functioning test, 4.6.1.3, and current capacity and circuit resistance after functioning test, 4.6.1.4.

4.4 Test Equipment and test conditions.

4.4.1 Quality conformance inspection equipment. The following items of test equipment are required to perform the quality conformance inspection tests.

	Equipment	Capability
a.	Millivoltmeter	Measure 0.00 to 150 millivolts DC; accuracy plus or minus 2 percent. Input impedance shall not be less than 1000 ohms.

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	Equipment	Capability
b.	DC power supply, current regulated	Regulate direct current at 2.0 plus 0.0 minus 0.2 amperes through switch contacts.
c.	Megohmmeter	Measure 0 to 50 megohms; accuracy 5 percent minimum; 500 plus or minus 50 volts DC potential.
d.	Centrifuge	Accelerate fixture of safety switch tester from 0 to 20g; accuracy 3 percent; minimum of 11 slip rings to provide connection from test fixture to control panel.
e.	Time interval counter	Measure intervals in milliseconds; accuracy plus or minus 2 percent.
f.	Ammeter	Measure 0.0 to 3.0 amperes DC; accuracy plus or minus 3 percent.
g.	Safety Switch Acceptance Tester	Conform to LD 624603; operate in accordance with OD 13065.

4.4.2 Preproduction and periodic production test equipment. The following items of test equipment, in addition to selected items of quality conformance inspection equipment and pieces of equipment recommended by applicable specifications or standards referenced within test procedure paragraphs, are required to perform the preproduction and periodic production tests:

	Equipment	Capability
a.	Temperature chamber	Maintain temperature at: Minus 65 plus 5 minus 0 degrees F Plus 35 plus 5 minus 0 degrees F Plus 120 plus 0 minus 5 degrees F Plus 160 plus 0 minus 5 degrees F
b.	Vibration testing machine	Control frequency from 20 cps to 2000 cps while subjecting the test item to 0.10-inch double amplitude or 10g acceleration, whichever is less; maintain direction of vibration along specified axes.

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Equipment	Capability
c. Shock equipment, standby	Apply shocks of 40 plus or minus 4g; a one-half sine wave pulse shape having a duration of 5 plus 3 minus 0 milliseconds: *maintain direction along specified axes.*

4.4.3 Test conditions. Unless otherwise specified, the switch shall be subjected to the preproduction and periodic production tests and quality conformance inspection tests under the following conditions:

a. Temperature	Room ambient (65 to 95 degrees F)
b. Altitude	Normal ground atmospheric pressure
c. Humidity (Relative)	Room ambient up to 95 percent

4.5 Preproduction and periodic production inspections. The preproduction and periodic production samples shall be subjected to the tests specified in 4.5.1 after satisfactorily passing the quality conformance inspections of 4.6.

4.5.1 Test procedure. Subject the switch to the test sequence in Table I. *Unless otherwise specified in the individual test paragraph,* the quality conformance inspection, as specified in Table I shall be performed subsequent to environmental tests under test conditions specified in 4.4.3. Failure of the switch to pass any one of these tests shall constitute a defective unit.

4.5.1.1 Temperature, storage condition. The switch, enclosed in a container with sufficient desiccant to prevent condensation of moisture and formation of frost on the switch, shall be subjected to a temperature of minus 65 plus 5 minus 0 degrees F for not less than 48 hours. The temperature shall be raised to plus 160 plus 0 minus 5 degrees F in not less than 2 hours. When the switch has reached room temperature during this 2-hour interval, remove the switch from the container. The temperature shall be maintained at plus 160 plus 0 minus 5 degrees F for not less than 48 hours. The unit shall then be subjected to room temperature for not less than 4 hours, prior to performing subsequent tests.

4.5.1.2 Vibration, standby condition. The switch, rigidly mounted as shown in Figure 2, shall be subjected to the vibration tests for Type I equipment, as specified in MIL-STD-167, up to 25 cps in each of the three axes shown in Figure 1.

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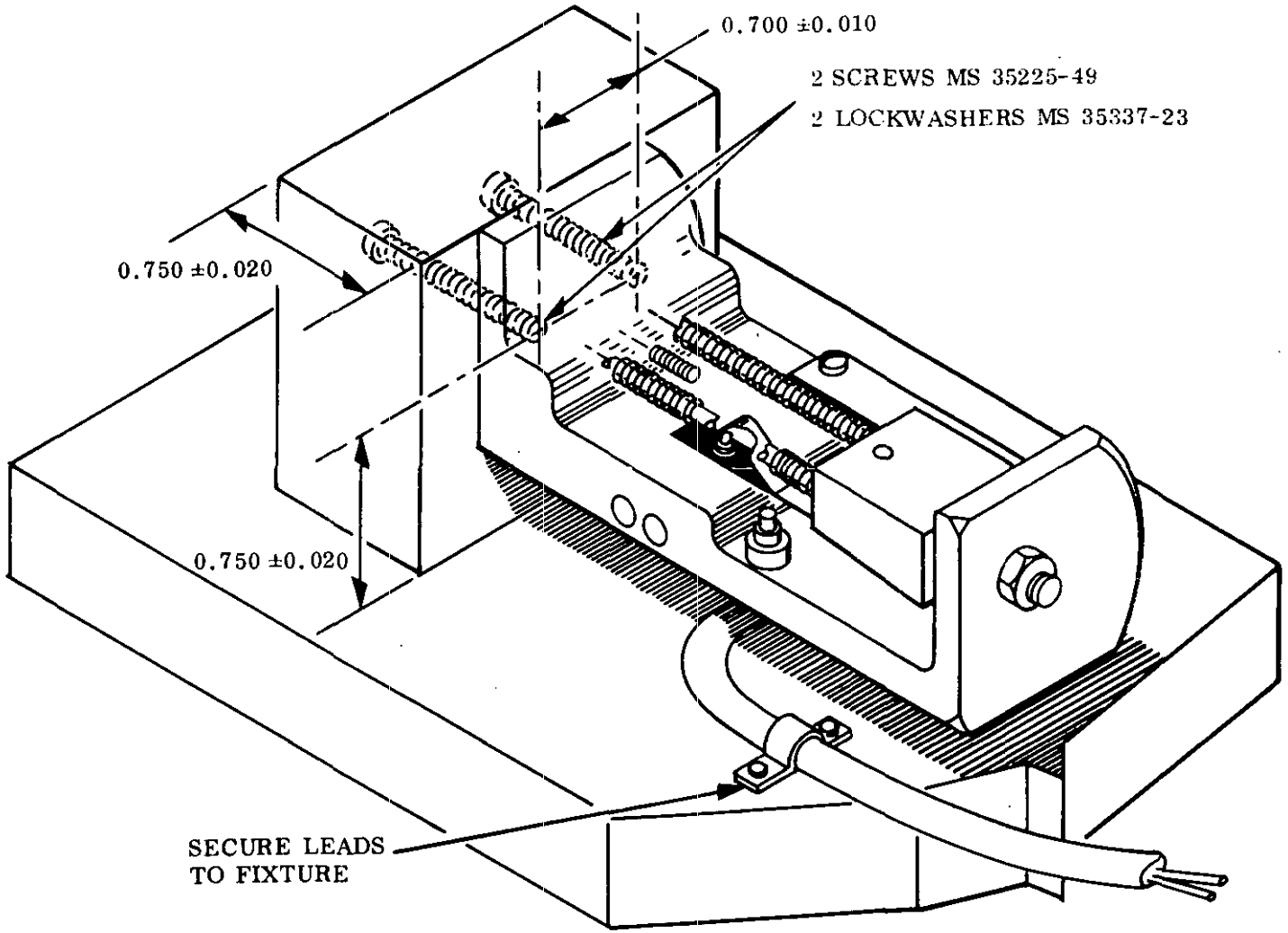


Figure 2. Safety Switch Mounting for Shock and Vibration

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4.5.1.3 Shock, standby condition. The switch, rigidly mounted as shown in Figure 2, shall be subjected to shocks of 40 plus or minus 4g peak magnitude with a pulse shape approximating a one-half sine wave having a duration of 5 plus 3 minus 0 milliseconds. Shocks shall be applied three times in each direction along each of three mutually perpendicular axes as shown in Figure 1.

4.5.1.4 Temperature, operating condition.

4.5.1.4.1 Low extreme. The switch, enclosed in a dry nitrogen atmosphere, shall be subjected to a temperature of 35 plus 5 minus 0 degrees F for not less than 2 hours. The switch, while exposed to this dry atmosphere and temperature, shall pass the quality conformance inspection tests.

4.5.1.4.2 High extreme. Subject the switch to a temperature of 120 plus 0 minus 5 degrees F for not less than 2 hours and while at this temperature, the switch shall pass the quality conformance inspection tests.

4.5.1.5 Vibration, operating condition. This test shall be performed after all other environmental tests have been completed. The switch shall be placed in the closed condition and shall be mounted as shown in Figure 2. Subject the switch to vibration from 20 to 2000 cps at 0.10-inch double amplitude or 10g, whichever is less, along each of three mutually perpendicular axes. The frequency shall be increased uniformly from 20 to 2000 cps. The frequency shall be traversed in not less than 2.5 minutes, once in each axis. During this period the current capacity and circuit resistance after functioning test of 4.6.1.4 shall be conducted.

4.6 Quality conformance inspection. Samples selected using the plans and procedures set forth in Table II of this specification, shall be subjected to the tests and examinations as described in 4.6.1.

Table II. Quality Conformance Tests and Examinations and AQL

Test	Sampling Plan	AQL	Test Paragraph
Insulation resistance tests	*	1.0	4.6.1.1
Acceleration sensitivity	*	0.25	4.6.1.2
Functioning tests	*	0.25	4.6.1.3
Current capacity and circuit resistance, after functioning	*	1.0	4.6.1.4

* MIL-STD-105D Inspection Level II

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Table II. Quality Conformance Tests and Examinations and AQL (continued)

Test	Sampling Plan	AQL	Test Paragraph
Visual examination	**	**	4.6.1.5
Preservation, packaging, packing, and marking	**	**	4.7

**One-hundred percent inspection; defect is cause for rejection of individual unit.

4.6.1 Test procedure. The switch shall be subjected to the tests and examinations as outlined herein. Unless otherwise specified, inability of the switch to pass the tests and examinations shall constitute a failure.

4.6.1.1 Insulation resistance test. Apply a potential of 500 plus or minus 50 volts DC between each lead, and between each lead and the case of the switch. Measure the resistance, using the megohmmeter specified in 4.4.1c. The resistance shall be not less than 20 megohms.

4.6.1.2 Acceleration sensitivity. Subject the switch, using the centrifuge specified in 4.4.1d, and the safety switch acceptance tester, specified in 4.4.1g, to an acceleration of 5 plus 0 minus 0.2g in the sensitive axis for not less than 30 seconds. (See Figure 1.) The normally open switch circuit shall not close.

4.6.1.3 Functioning tests. Subject the switch to the acceleration levels listed below, using the centrifuge specified in 4.4.1d and the safety switch acceptance tester, specified in 4.4.1g. Using the time interval counter, specified in 4.4.1e, measure the time required for the timing weight to leave the stop and close the switch circuit. The time interval for each level of acceleration shall be as specified below. Perform the tests three times at each acceleration level. Verify switch latching as described in 4.6.1.3.1

	Acceleration, g	Time, Seconds
a.	10 ±0.30	0.6 ±0.10
b.	15 ±0.45	0.4 ±0.10

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4.6.1.3.1 Switch latching. Following the functioning of the switch and using the safety switch acceptance tester, specified in 4.4.1g, rotate the sensitive axis of the switch out of the line of acceleration. The safety switch acceptance tester shall indicate that the switch remains in the closed condition.

4.6.1.4 Current capacity and circuit resistance after functioning. Apply a current of 2.0 plus 0.0 minus 0.2 amperes to the normally open circuit when closed, using the power supply specified in 4.4.1b. Verify correct current with the ammeter, specified in 4.4.1f. Measure the voltage drop across the leads at least 30 seconds after the application of current, using the millivoltmeter, specified in 4.4.1a. The voltage drop shall not exceed 125 millivolts.

4.6.1.5 Visual examination. Visual examination shall be made on each switch submitted for delivery to the procuring activity. The switch shall be examined for conformance to the requirements of good workmanship of 3.5. Nonconformance is cause for rejection of individual unit.

4.7 Preservation, packaging, packing, and marking. The switch shall be examined for conformance to the prescribed preservation, packaging, packing, and marking requirements of Section 5.

NOTE: Upon completion of the quality conformance inspection tests, and prior to packaging, the switch shall be protected from a humidity environment greater than 35 percent relative.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified. (See 6.2b and 6.4).

5.1.1 Level A

5.1.1.1 Cleaning. The switch shall be cleaned in accordance with Method C-1, MIL-P-116.

5.1.1.2 Drying. The switch shall be dried in accordance with Method D-1, MIL-P-116.

5.1.1.3 Unit packaging. The switch shall be individually packaged in accordance with Method IIb of MIL-P-116.

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5.1.2 Level C. Cleaning, drying, preservation, and packaging shall be in accordance with the supplier's commercial practice, when such practices have been approved by the procuring activity.

5.2 Packing.

5.2.1 Level A. Unless otherwise specified in the contract, order, or requisition, the switch shall be packed in exterior type containers, PPP-B-621, of uniform size and shape, with minimum cube and tare consistent with the protection required. The gross weight of each pack shall be limited to approximately 100 pounds. Containers shall be provided with a case liner, MIL-L-10547, and shall be closed and strapped in accordance with PPP-B-621.

5.2.2 Level C. The switch packaged as specified in 5.1.2 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129. Unit packages shall be marked or tagged with the words DO NOT OPEN UNTIL READY FOR USE.

6. NOTES

6.1 Intended use and description.

6.1.1 Intended use. The switch is intended to be used as a circuit closing device in a rocket propelled missile.

6.1.2 Description. The safety switch is designed to maintain an electrical circuit open until a constant acceleration of 8g or more is experienced.

6.2 Ordering data. Procurement documents should specify the following:

a. Title number and date of this specification.

b. Required level of preservation and packaging.

c. Required level of packing.

d. Any exceptions to this specification.

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6.2.1 Additional samples. Additional samples required because of failure of any sample(s) to pass the prescribed inspections shall be provided and inspected at the expense of the supplier.

6.3 Definitions. For the purpose of this specification the following definitions have been established.

6.3.1 Unit condition.

6.3.1.1 Operating. The condition of the switch during missile operation.

6.3.1.2 Standby. Standby is the condition of the switch when aboard ship and is assembled into the end item.

6.3.2 Sensitive axis. The sensitive axis is the longitudinal axis along which the weight moves to actuate the switch when this axis is oriented in the direction of the acceleration.

6.4 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as specified in FED-STD-102.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-S-82519(OS) SWITCH, SAFETY MARK 61 MOD 1

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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